MIL-T-23592(Wep)
23 January 1963

MILITARY SPECIFICATION

FOR

TEST SET, INFRARED DETECTING SET AN/AAM-6

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for flight line and shop use, and shall be capable of fault isolation and checkout of the Detecting Set, Infrared AN/AAS-15.

1.2 Classification - The Test Set, Infrared Detecting Set AN/AAM-6 shall consist of the following items:

Items	Type Designation	Applicable Para.
Test Set, Infrared	TS-1839/AAM-6	3.4.1
Case, Transit	CY-3745/AAM-6	3.4.2
Interconnecting Cable Assy.	W-1	3.4.3

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

MIL-C-4150	Case, Carrying and Storage, Shock and Waterproof
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of

FSC 1290

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MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-L-10547	Liners, Case, Waterproof
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-P-116	Preservation, Methods of
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-T-21200 (ASG)	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-W-5088 (ASG)	Wiring, Aircraft, Installation of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-242	Electronic Equipment Parts
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200 Volt 3-Phase AC, or Combinations
MS 28053	Meter, Time Totalizing, 400-Cycle

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PUBLICATIONS

Bureau of Naval Weapons

SAR-300 Material Changes and Material Bulletins, Preparation of

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Ave., Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

(2) Nonrepairable subassemblies, as outlined in specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable sub-assemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

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3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 75 pounds.

3.2.3 Operating Life

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during normal use.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 Mean Operating Time Between Failures - The equipment shall have a minimum of 200 hours of mean operating time between failures. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.

3.2.4 Cabling and Connections

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200.

3.2.4.2 Interconnection Cablings - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall provide shielded wires where necessary to meet interference control requirements provided the assembly of the cable to its plugs may be easily accomplished.

3.2.4.3 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

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3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance specified in Specification MIL-T-21200 shall apply. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (30° ±10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ±1.0 V AC 400 cycles and 27.5 ±0.5 V DC

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combintaion of these conditions as specified in Specification MIL-T-21200 for Class 2 equipment.

3.2.11 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

3.2.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-T-21200 for Class 2 equipment. The power required shall be of the following type and shall not exceed the specified amounts.

AC Power, Single Phase, 115V, 400 cps, 80 VA

3.2.12.1 Voltage Protection - The equipment shall not be damaged by voltages either above or below the voltages specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.13 Time Totalizing Meter - A time totalizing meter conforming with Standard MS 28053 shall be included.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

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3.3.2 Test Capabilities - The Test Set, Infrared, in conjunction with an IR Target Simulator, AN/AAM-5, when an IR source is required and suitable oscilloscope, shall be capabile of checking the following functions and circuits:

a. Video Amplifiers (Lin-Log)

b. Receiver Detectors and Preamplifiers. (Checked indirectly by logical fault isolation between IR Receiver and video amplifiers.)

c. Receiver Reference Generator

d. Reference Signal Buffer Amplifiers

e. Sample and Hold Circuitry

f. Slaving Phase Detectors

g. Track Loop

h. Acquisition Loop

i. Four Bar Stepping Circuitry

3.3.3 Test Signal Characteristics - The Test Set, Infrared, shall provide test signals with the following characteristics:

3.3.3.1 Acquisition and Track Servo Test Signals -

a. Track - 4.0V P to P ± 10% 1/2 cps ± 10% 1.0 meg load
4V P to P ± 10% 2 cps ± 10% 1.0 meg load

b. Acquisition -
4.0V P to P ± 10% 2 cps ± 10% 1.0 meg load
4.0V P to P ± 10% 10 cps ± 10% 1.0 meg load
4.0V P to P ± 10% 24 cps ± 10% 1.0 meg load

3.3.3.2 Azimuth and Elevation Buffer Amplifier Test Signal -

a. 10 volts P to P ± 10%, 45 cps ± 10% 5K Load

3.3.3.3 Slaving Phase Detector Test Signals -

a. 0.0 V

b. +1V rms ± 5%, 400 ± 20 cps

c. -1V rms ±5%, 400 ± 20 cps

d. +3V rms ± 5%, 400 ± 20 cps

e. -3V rms ± 5%, 400 ± 20 cps

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3.3.3.4 Pulse Test Signals -

a. Signal levels

1) 0.0
2) 0.1 V ± 10%
3) 0.4 V ± 10%
4) 1.6 V ± 10%

b. Relative variation of signal levels ± 2%

c. Pulse width 600 ± 100 microseconds

d. Phase - The phase of the pulse shall be capable of being shifted 90° and 270° with respect to the 45 cps sine wave.

3.3.4 Indicators - The Test Set, Infrared shall be capable of reading voltages directly without the use of external equipment and capable of having an AN/USM-105A or equivalent used for monitoring various voltages.

3.3.4.1 VTVM - The VTVM shall have the following ranges with a full scale accuracy of ± 5%.

(1) +20 VDC

(2) -10 to +10 VDC (center zero)

(3) -2.94 to +2.94 VDC (center zero)

3.4 Detail Requirements -

3.4.1 Test Set, Infrared TS-1839/AAM-6 - The Test Set, Infrared shall meet the following requirements:

3.4.1.1 Form Factor - The Test Set, Infrared, will not exceed 15 by 10 by 18.38 inches.

3.4.1.2 Weight - The weight of the Test Set, Infrared shall not exceed 45 pounds.

3.4.1.3 Contents of Unit - The Test Set, Infrared shall contain the following assemblies, subassemblies, circuits and panels:

(1) Front Panel - The front panel shall mount test points, switches, lights, and a roll chart.

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(2) Rear Panel - The rear panel shall mount a signal connector, power connector, running time meter, and a d.c. zero adjustment for the operational amplifier.

(3) Oscillator, Electrical - Modular Assembly - The oscillator electrical modular assembly shall provide sinusoidal outputs in accordance with section 3.3.3.1 and 3.3.3.2. A Wien bridge oscillator shall be used.

(4) Power Supply, Electrical - Modular Assembly - The power supply electrical modular assembly shall provide +200V ± 10%, -150V ±10% +23.4V ± 6%, -23.4V ± 6%, +32V ± 6%, +18V ± 6%, -59V ± 6% and -22V ± 6% under full load conditions internal to the test set. Full load shall be defined as the current normally drawn by the test set circuitry. The +200 and -150 voltages may be unregulated. All other voltages shall be regulated by zener diodes. (The voltages in this assembly do not appear at the output connector.)

(5) Pulse Generator - Modular Assembly - The pulse generator modular assembly shall provide pulses in accordance with section 3.3.3.4.

(6) VTVM - Modular Assembly - The VTVM shall have voltage ranges as specified in 3.3.4.1.

(7) Operational Amplifiers - Two operational amplifiers, each having a gain of minus one, shall be switched into the acquisition and track loops to check servo response.

(8) IR Receiver Positioning Circuit - The IR receiver positioning circuit shall set the system in the Acquisition mode and provide a means of accurately positioning and holding the IR receiver dead ahead.

(9) Roll Chart - The roll chart shall be accessible from the front of the unit and shall contain the procedure and tolerances necessary to fault isolate and check out the Infrared Detecting Set AN/AAS-15.

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3.4.1.4 Controls - The Test Set, Infrared shall have the following controls:

(1) POWER SWITCH - A POWER ON-OFF switch shall be located on the front panel. This switch shall also provide circuit protection. The switch shall allow 115V 400 cycle power to be applied to the Test Set, Infrared.

(2) SELECTOR - A SELECTOR switch shall be located on the front panel to provide selection of the following for test or monitoring:

1. LIN LOG VID
2. REF BUFF AMPL 1
3. 01 — TRACK ERR
4. 02 — TRACK ERR
5. 03 — TRACK ERR
6. P1 — LOG SIG STR & VID
7. P2 — LOG SIG STR & VID
8. P3 — LOG SIG STR & VID
9. P4 — LOG SIG STR & VID
10. REF BUFF AMPL 2
11. NULL — SLAVING P. D.
12. + — SLAVING P. D.
13. - — SLAVING P. D.
14. + — SLAVING P. D.
15. - — SLAVING P. D.
16. F1 — TRACK ERR
17. F2 — TRACK ERR
18. F1 — ACQ ERR
19. F2 — ACQ ERR
20. F3 — ACQ ERR
21. 1 & 3 — FOUR BAR STEP
22. 2 & 4 — FOUR BAR STEP

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(3) CHANNEL SELECT - A CHANNEL SELECT shall be located on the front panel for the selection of azimuth or elevation channels when required in testing.

(4) VTVM PUSH-TO-ZERO - A VTVM PUSH-TO-ZERO push button shall be located on the front panel in order to zero the VTVM.

(5) VTVM ZERO ADJ - A VTVM ZERO ADJ control knob shall be located on the front panel in order to zero the VTVM.

(6) ROLL CHART CRANK - A ROLL CHART CRANK shall be located on the front panel.

The crank will allow the selection of various instruments printed on the roll chart located at the front panel.

(7) OPERATIONAL AMPL. INPUT SWITCH - An OPERATIONAL AMPL. EXTERNAL INPUT SWITCH shall be located on the front panel. When this switch is in the "INTERNAL" position the operational amplifiers (ref. 3.4.1.3, item 7) shall receive an input signal from the oscillator module (ref. 3.4.1.3, item 3). When in the "EXTERNAL' position, this switch will allow the operational amplifiers to receive an input signal from an external source.

(8) OPERATIONAL AMPL DC ADJ - An OP AMPL DC ZERO screw-driver adjustment shall be provided on the rear panel to compensate for amplifier drift.

(9) IR RECEIVER POSITIONING SWITCH - An IR REC POS SW shall be located on the front panel to disconnect the IR receiver positioning circuitry when not in use.

(10) IR RECEIVER POSITIONING POTENTIOMETERS - Two potentiometers shall be located on the front panel to accurately position and hold the IR receiver dead ahead.

3.4.1.5 Electrical Connections - Connections to external circuits shall be provided as follows:

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3.4.1.5.1 Front Panel Test Jacks - The following test jacks shall be provided on the front panel:

Reference Designation	Function
J 90001	SYSTEM TEST JACK
J 3010	INTENSITY VIDEO (Test Point)
J 3009	DEFL VIDEO (Test Point)
J 3008	SLAVING P.D. OUT (Test Point)
J 3003	PULSE OUT (Test Point)
J 3011	θ IN (Test Point)
J 3012	θ OUT (Test Point)
J 3013	θ ERROR (Test Point)
J 3004	OSC OUT (Test Point)
J 3007	AURAL TONE (Test Point)
J 3005	CHASSIS GRD (Test Point)
J 3006	SYSTEM GRD (Test Point)
J 3001	SCOPE JACK

3.4.1.5.2 Internal Test Points - The following internal test points shall be provided:

Reference Designation	Function
J 3102	TRIG INPUT
J 3103	TRIG OUT
J 3104	MVB OUT
J 3202	+200V
J 3203	+32V
J 3204	+18V

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Reference Designation	Function
J 3205	-.75 REF
J 3206	DC RTN
J 3207	+23.4V
J 3208	-22V
J 3209	-150V
J 3210	-59V
J 3211	-23.4V
J 3302	OSC OUT

3.4.2 Case, Transit CY-3745/AAM-6 - The transit case for the Test Set shall be in accordance with Specification MIL-C-4150, Size VII.

3.4.2.1 Form Factor - The dimensions of the transit case shall not exceed 21 by 20 by 19 inches.

3.4.2.2 Weight - The weight of the transit case shall not exceed 30 pounds.

3.4.3 Interconnecting Cable Assembly, W-1 - The interconnecting cable assembly shall meet the applicable requirements of MIL-T-21200.

3.4.3.1 Function - The interconnecting cable assembly shall provide the interconnection of the Test Set, Infrared with the AN/AAS-15 Detecting Set, Infrared.

3.4.3.2 Length - The interconnecting cable assembly W-1 shall be 10 feet + 1/2 ft.
- 0 ft.

3.4.3.3 Contents of Unit - The interconnecting cable assembly W-1 shall consist of the following:

(1) Connector - (DEUTSCH 17942-61S-395) P90001 - The connector P90001 joins the cable assembly with the test set through jack J90001.

(2) Connector - (DEUTSCH 17942-61P-395) 2P1 - The connector 2P1 joins the cable assembly to the AN/AAS-15 through the jack 2J1.

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(3) Cable - The cable consists of shielded and unshielded insulated stranded wires in a neoprene cover.

3.5 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

(1) AN/AAS-15 - Detecting Set, Infrared

(2) AN/AAM-5 - Target Simulator

(3) AN/USM-105A - Oscilloscope

(4) AN/AAM-4 - Test Bench Harness, Infrared Set

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable re-requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedures of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

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4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, including the Sand and Dust Test and modifying the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with Specification MIL-I-6181. Drop tests for transit and combination cases shall be in accordance with Specification MIL-T-21200.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. At the discretion of the procuring activity, no production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

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4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

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(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which cause malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment;

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 Minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions, the vertical direction should be used. During the 6-hour period of operation following the 10-minute vibraion period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing reliability test specified in Para. 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment taken.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

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(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

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4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-7936. For level A preservation and packaging of MIL-P-116 shall be Method II utilizing the transit case to enclose the desicant.

6. NOTES

6.1 Use - The Test Set, Infrared Detecting Set AN/AAM-6 is intended for use in checking out the AN/AAS-15 in the F8U-2N and F8U-2NE aircraft on the flight line, flight deck hanger and shop checkout and repair.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V and 115 ± 1.0 V, 400 cycles. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 The parentheses (*) when used in the type designations, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with Para. 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books as applicable.

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6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.6 NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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NAVY DEPARTMENT

Chief, Bureau of Naval Weapons (RREN-5)

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